

EX.no:04

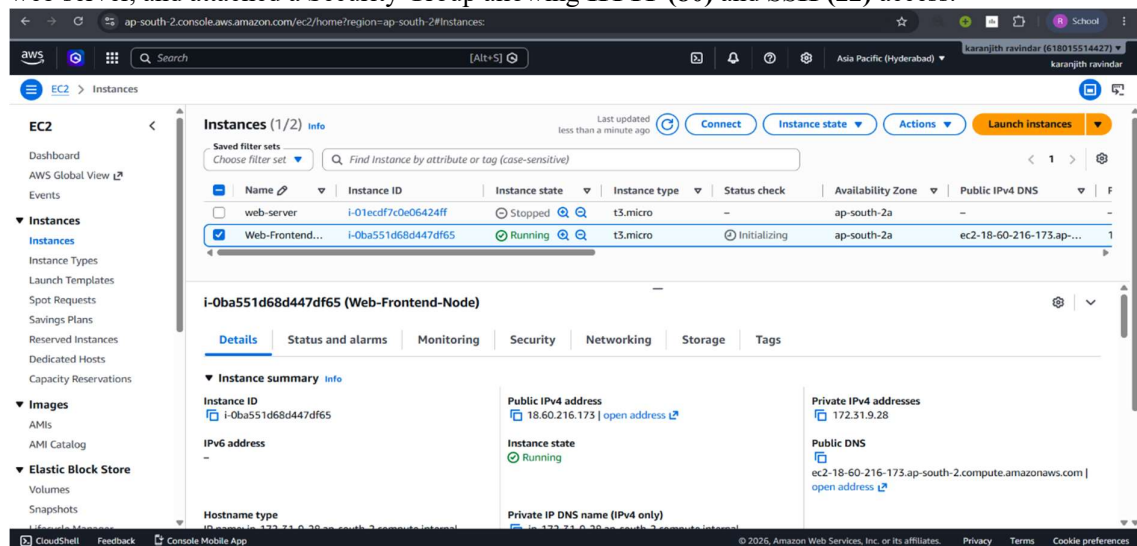
PRACTICE: AWS EC2 Security Group Isolation and Connectivity Verification

Aim

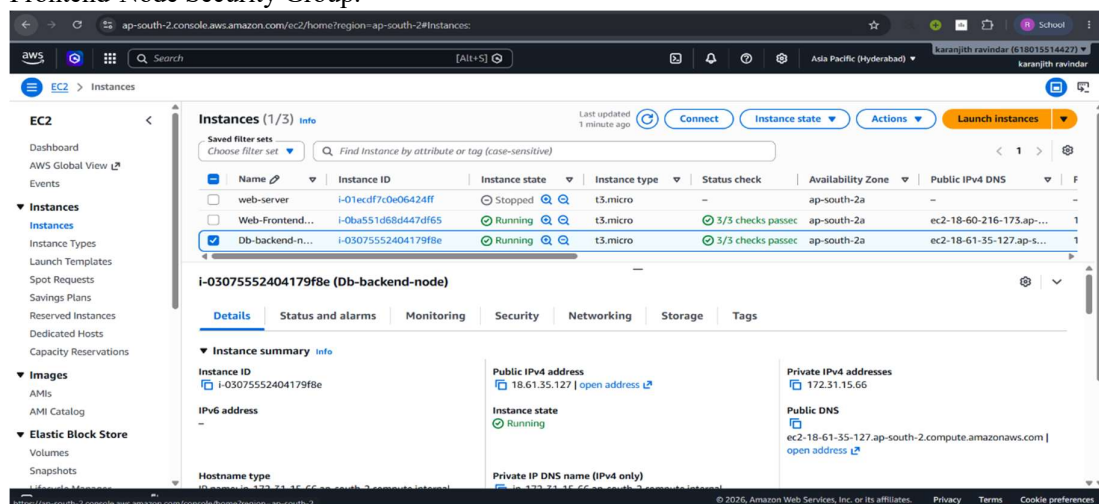
To deploy two EC2 instances, configure Security Groups for secure communication, and verify network isolation between the web server and database server.

Procedure

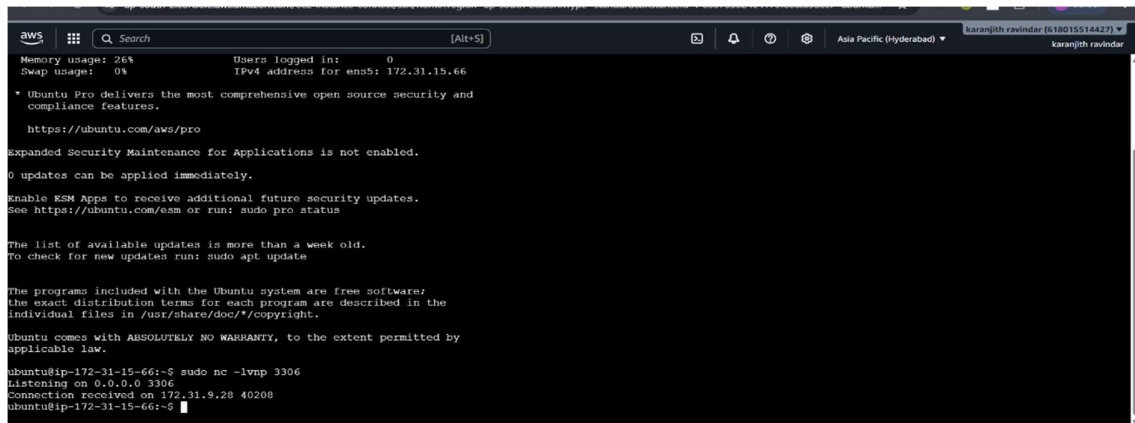
1. Launched an EC2 instance named **Web-Frontend-Node**, added a User Data script to install a web server, and attached a Security Group allowing **HTTP (80)** and **SSH (22)** access.



2. Launched another EC2 instance named **Database-Backend-Node** in the same Availability Zone and attached a **DB-Isolation-SG** allowing **MySQL (3306)** access only from the Web-Frontend-Node Security Group.



3. Connected to the **Web-Frontend-Node** using SSH.

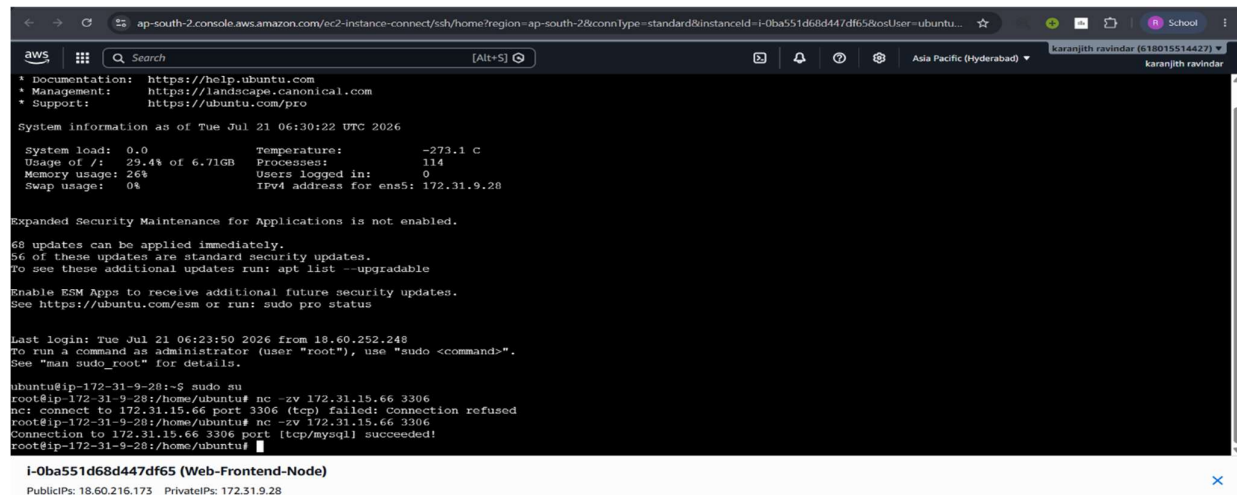


```
aws
Memory usage: 26%      Users logged in: 0
Swap usage: 0%        IPv4 address for ens5: 172.31.15.66
* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.
  https://ubuntu.com/aws/pro
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
ubuntu@ip-172-31-15-66:~$ sudo nc -lvp 3306
Listening on 0.0.0.0 3306
Connection received on 172.31.9.28 40208
ubuntu@ip-172-31-15-66:~$
```

4. Tested connectivity to the database instance using:

`nc -zv <Database_Private_IP> 3306`

and confirmed that the connection was successful.



```
aws
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Tue Jul 21 06:30:22 UTC 2026
System load: 0.0      Temperature: -273.1 C
Usage of /: 29.4% of 6.71GB    Processes: 114
Memory usage: 26%      Users logged in: 0
Swap usage: 0%        IPv4 address for ens5: 172.31.9.28
Expanded Security Maintenance for Applications is not enabled.
68 updates can be applied immediately.
56 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Tue Jul 21 06:23:50 2026 from 18.60.252.248
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-172-31-9-28:~$ sudo su
root@ip-172-31-9-28:/home/ubuntu# nc -zv 172.31.15.66 3306
nc: connect to 172.31.15.66 port 3306 (tcp) failed: Connection refused
root@ip-172-31-9-28:/home/ubuntu# nc -zv 172.31.15.66 3306
Connection to 172.31.15.66 3306 port: [tcp/mysql] succeeded!
root@ip-172-31-9-28:/home/ubuntu#
```

i-0ba551d68d447df65 (Web-Frontend-Node)
PublicIPs: 18.60.216.173 PrivateIPs: 172.31.9.28

5. Attempted to access the database instance directly from the public network and observed that the connection was blocked due to the Security Group restrictions.

Result

The Web-Frontend-Node successfully communicated with the Database-Backend-Node through its private IP, while direct public access to the database server was blocked, demonstrating effective Security Group isolation.